

SwiftRoute E-Commerce Shipping Data EDA

Practical Assessment Report - Dataset Acquisition and Exploratory Data Analysis

Dataset Loading and Quality Checks

The dataset was loaded from Kaggle's E-Commerce Shipping Data file. It contains 10,999 records and 12 columns. No missing values or duplicate records were found. The target variable is Reached.on.Time_Y.N, where 1 means not delivered on time and 0 means delivered on time.

	dtype
ID	int64
Warehouse_block	str
Mode_of_Shipment	str
Customer_care_calls	int64
Customer_rating	int64
Cost_of_the_Product	int64
Prior_purchases	int64
Product_importance	str
Gender	str
Discount_offered	int64
Weight_in_gms	int64
Reached.on.Time_Y.N	int64

	missing
ID	0
Warehouse_block	0
Mode_of_Shipment	0
Customer_care_calls	0
Customer_rating	0
Cost_of_the_Product	0
Prior_purchases	0
Product_importance	0
Gender	0
Discount_offered	0
Weight_in_gms	0
Reached.on.Time_Y.N	0

Target Class Balance

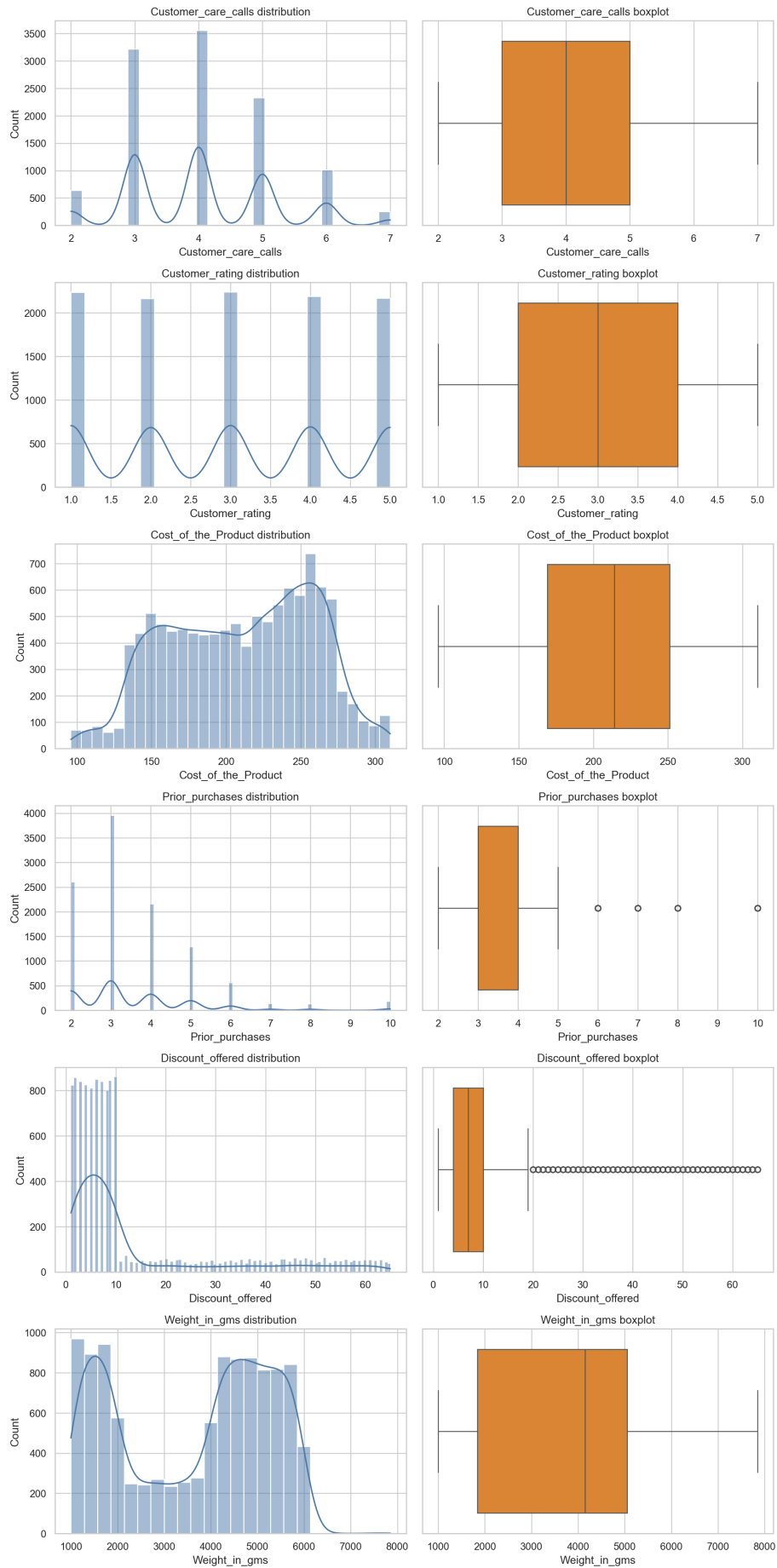
	count	percent
On time (0)	4436	40.33
Not on time (1)	6563	59.67



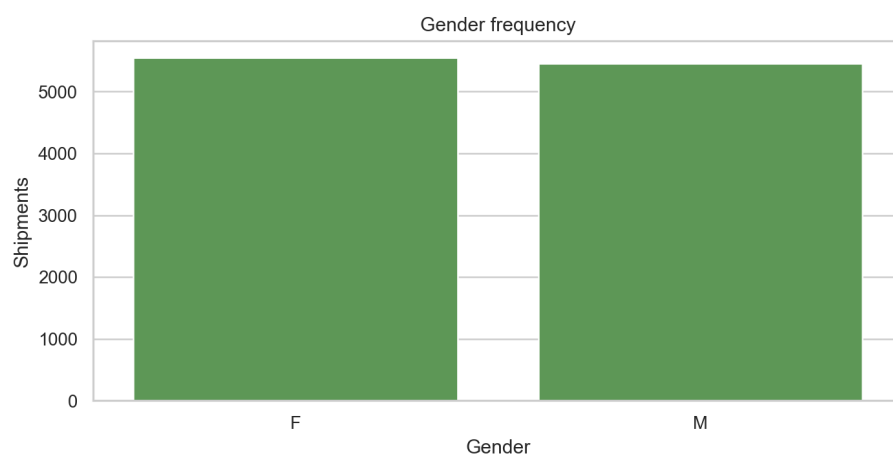
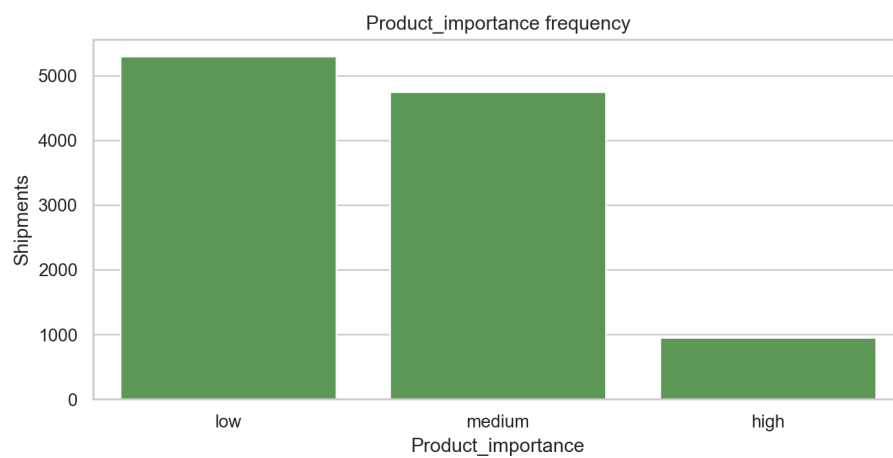
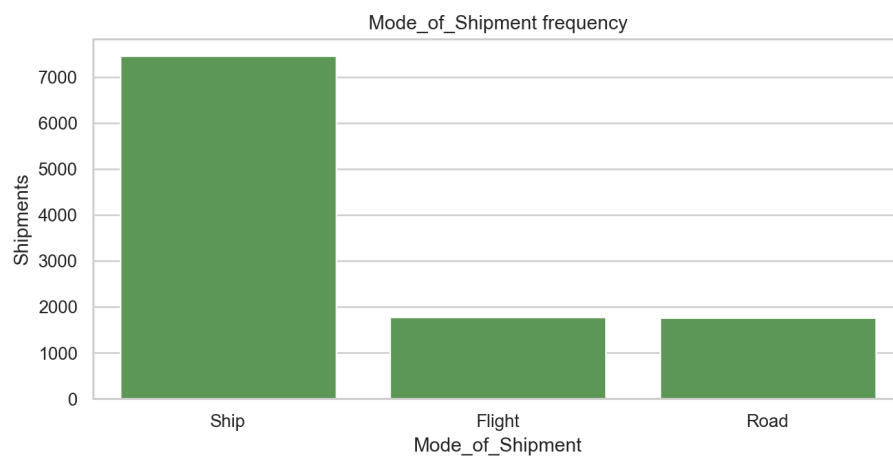
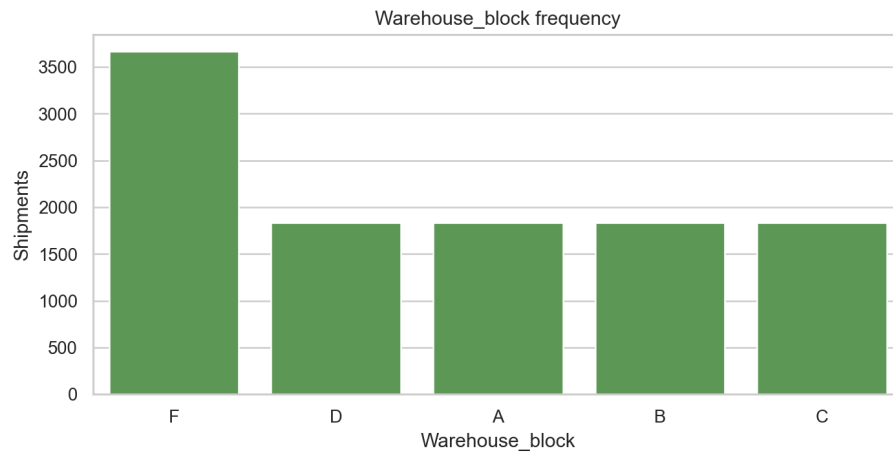
Univariate Analysis

Numeric variables include bounded count/rating fields and wider continuous measures. Discount offered is strongly right-skewed, while cost and weight show broad spreads. Categorical fields are clean, with shipment mode dominated by Ship and product importance mostly low or medium.

	count	mean	std	min	25%	50%	75%	max	skewness
Customer_care_calls	10999.0	4.05	1.14	2.0	3.0	4.0	5.0	7.0	0.39
Customer_rating	10999.0	2.99	1.41	1.0	2.0	3.0	4.0	5.0	0.0
Cost_of_the_Product	10999.0	210.2	48.06	96.0	169.0	214.0	251.0	310.0	-0.16
Prior_purchases	10999.0	3.57	1.52	2.0	3.0	3.0	4.0	10.0	1.68
Discount_offered	10999.0	13.37	16.21	1.0	4.0	7.0	10.0	65.0	1.8
Weight_in_gms	10999.0	3634.02	1635.38	1001.0	1839.5	4149.0	5050.0	7846.0	-0.25

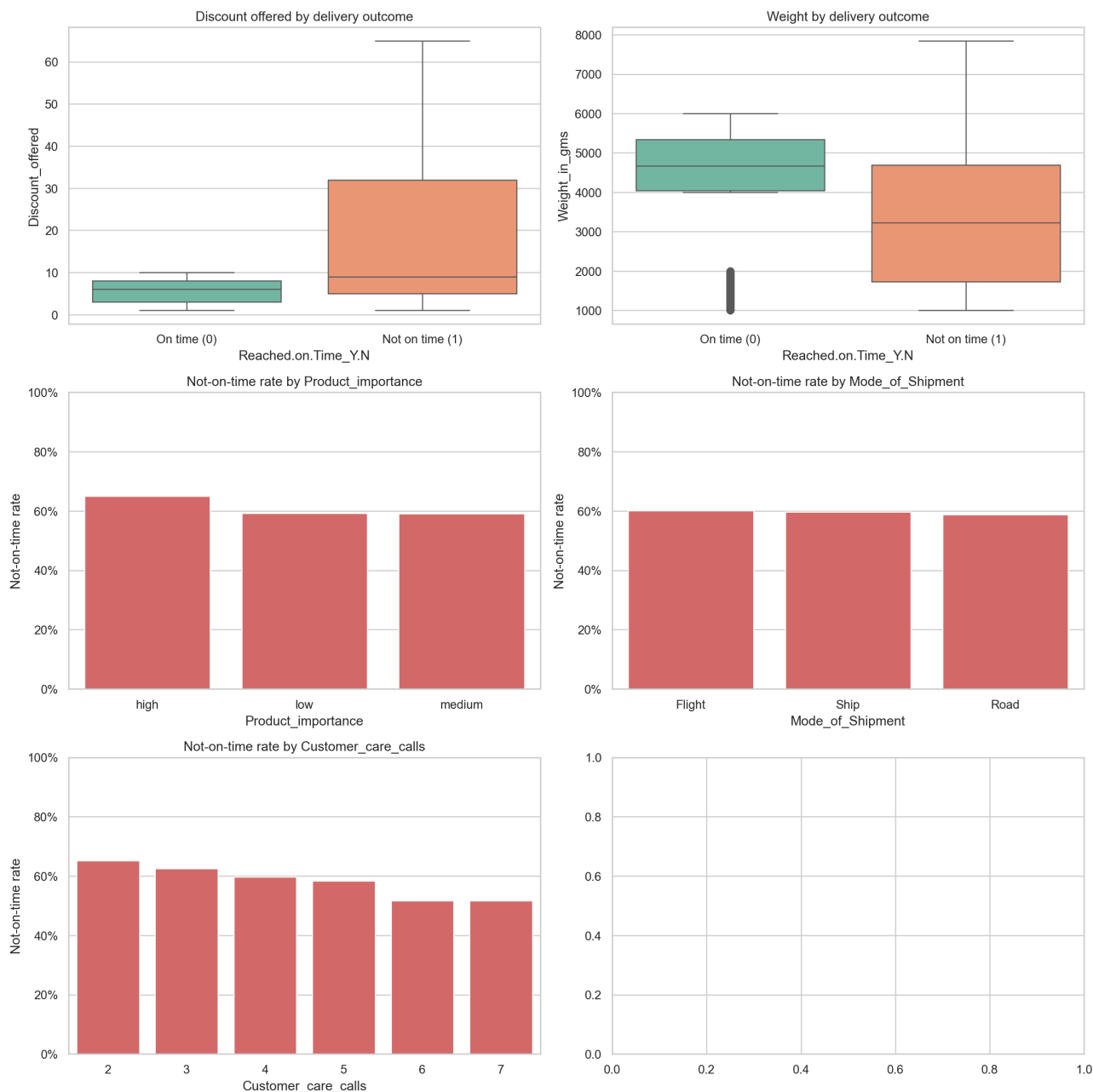


Categorical Feature Distributions



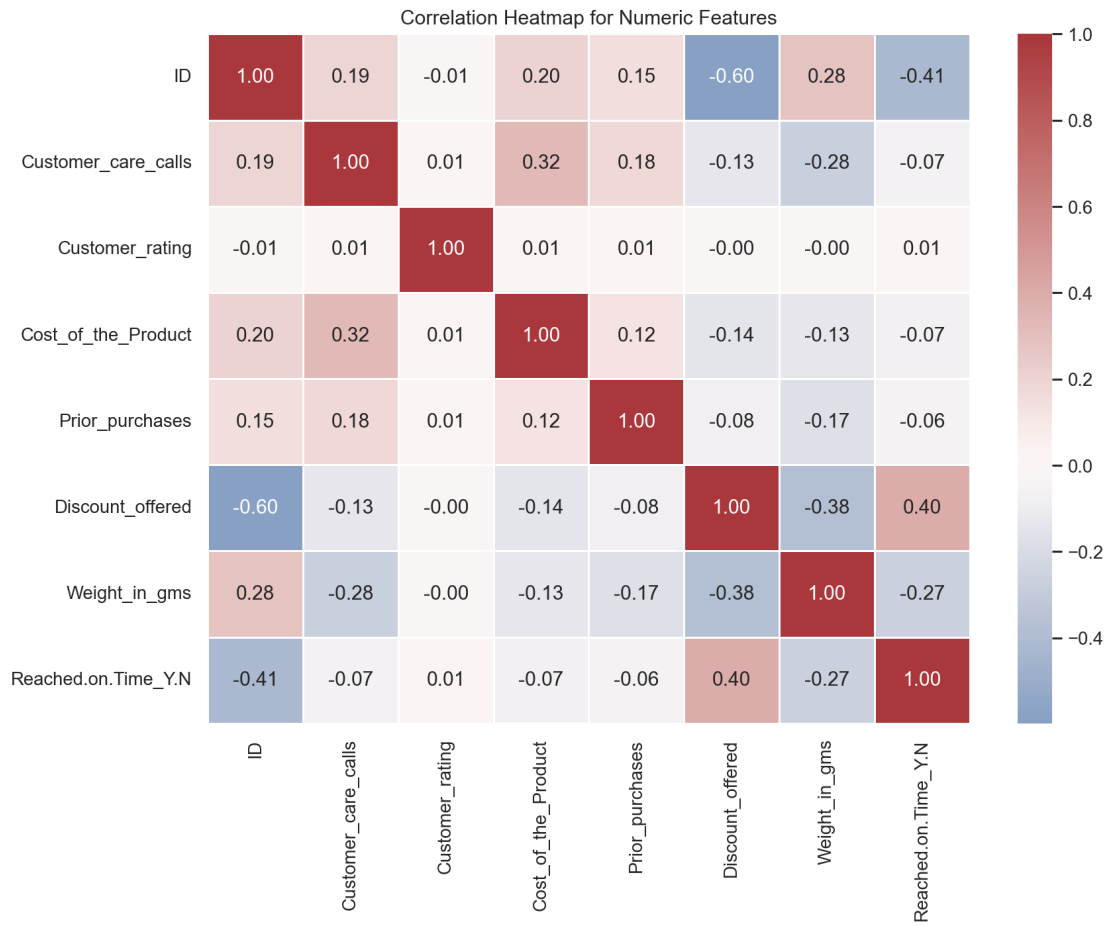
Bivariate Analysis Against Target

Discount offered and shipment weight show the clearest visual separation between the two target classes. Product importance, mode of shipment and customer-care calls also show useful differences in not-on-time rates.



Correlation Analysis

The strongest non-target numeric relationship is between Cost_of_the_Product and Customer_care_calls. The strongest target-related relationship is between Discount_offered and Reached.on.Time_Y.N.



Written Findings Report

The E-Commerce Shipping dataset contains 10,999 shipment records and 12 fields, with no missing values and no duplicate rows. The target variable is imbalanced but still usable for modelling: 6,563 shipments (59.67%) are labelled 1, meaning not delivered on time, while 4,436 shipments (40.33%) are labelled 0, meaning delivered on time. The imbalance is moderate, so evaluation should not rely on accuracy alone; precision, recall, F1-score and a confusion matrix would be more informative.

The univariate plots show that several variables are discrete and bounded, including customer-care calls, customer rating and prior purchases. Customer ratings are fairly spread across the 1 to 5 scale, while prior purchases are concentrated at lower values with a few high observations. Discount offered is the most skewed numeric feature, with many small discounts and a smaller number of very high discounts. Cost of the product and shipment weight have wider spreads, and the boxplots suggest possible outliers in discount and prior purchases. Categorical fields appear clean: warehouse block F has the largest share, shipment mode is dominated by Ship, and product importance is mostly low or medium.

The bivariate analysis suggests that Discount_offered is likely to be one of the most predictive features because not-on-time shipments have visibly higher discount values. Weight_in_gms also looks important, as the two target classes show different weight distributions, with on-time deliveries tending to be heavier. Customer_care_calls and Product_importance may add useful signal because their not-on-time rates vary by group, although their relationships are weaker than discount and weight. Mode_of_Shipment shows some variation, but the dominance of Ship means it should be interpreted carefully. The correlation heatmap supports these findings: Discount_offered has the strongest numeric association with the target, while Cost_of_the_Product and Customer_care_calls form the strongest non-target relationship. ID is a unique identifier and should be excluded from predictive modelling because it is not a meaningful shipment attribute.

Word count: 309

Repository Submission Contents

The GitHub-ready package includes the dataset, reproducible source code, report files, figures and CSV summaries. Later parts covering Logic and Knowledge, Search and CSP, and broader AI Foundations should be added once their full prompts are provided.